The eight-study portfolio: the question each study asks

grouped by theme; the question is the study's own and the status is the state of the evidence in this thesis

Scaling

does the recipe follow a law?

P1 · P3

P1 **Scaling laws for GRPO post-training**

Does held-out GRPO gain follow a predictable cross-scale slope, or does the recipe resist a scaling law?

partial · no reliable training-reward scaling trend; cross-scale held-out gain remains untested

P3 **Group size, contrast, DPO bridge**

How do collapse, reward, held-out gain and token cost move with group size G — and what does G buy?

complete · under saturation, training-reward ratio ≈ 1.00 across G (equivalence to G=2 through G=8); SNR is 52% of the $\sqrt{G}$ ideal

Diagnostic

what is the signal doing?

P2

P2 **The Zero-Variance Fraction as a descriptive diagnostic**

What fraction of each rollout batch supplies no reward-relative gradient, and can a collapsed group be re-baselined?

complete · recomputable from stored reward tensors (0.130, 0.190, 0.155; pooled 0.158); held-out predictive validity unestablished

Mechanism

why does the signal vanish?

P4 · P7

P4 **Length bias and held-out generalisation**

Do length penalties change how long a model answers — and does that change held-out accuracy at all?

complete · held-out gains indistinguishable; pooled length 194.4 → 183.5 tokens under a 200-token cap

P7 **From diagnostic to controller**

Can the ZVF diagnostic be closed into a controller that adapts group size from measured starvation?

partial · 368-run audit; N10 changes the decision rule; no P5 integration

Reporting

what must be stated?

P5 · P6

P5 **Report the stack, not the label**

What minimum set of fields must a stack-conditioned result report before it can be attributed at all?

complete · flagship: the seven-field minimum-report manifest; badge 43–96 over twenty seed entries

P6 **GRPO-Registry: a stack catalogue**

Can evolving run state be recorded as a machine-readable registry, and diffed to expose declared

complete · *stackdiff* assigns the open-vs-closed DAPO pair declared-difference code R5 from manifests alone (schema proposal; not a calibrated flip probability)

Applied

does it transfer?

P8

P8 **LLM vs. gradient-boosted trees in credit-card fraud**

When two models are measured on different splits, what can the comparison actually support?

partial · archived side study absorbed as a measurement-discipline case study; the two AUCs are not like-for-like

complete — the study answers its question, including when the answer is a null

partial — scope narrowed or a limb deliberately left unclaimed

Reading the map

Established

- Stored reward tensors reproduce ZVF: 0.130, 0.190, 0.155.
- Under saturation, reward-ratio equivalence to G=2 holds through G=8, not G=16 (p=0.056).
- Seven-field manifests expose declared differences (schema proposal).

Declared null

- P1 recovers no reliable cross-scale slope and no identifiable saturation exponent.
- P4 finds GRPO and Dr. GRPO held-out gains indistinguishable.
- Its error-bar audit, re-graded, returns no decisive verdict and one suggestive.

Not claimed

- P7 is audited; joint attribution with P5 remains untested.
- P8's AUCs use different splits and are not a ranking.
- P2 mixes synthetic cells and outcome definitions; held-out prediction remains unvalidated.